

Food Ordering Behaviour and Consumer Trends Project Report

ABSTRACT

Food Ordering Behaviour and Consumer Trends is a Business Intelligence project that analyzes consumer food choices, ordering habits, and influencing factors using Tableau. A food delivery dataset from Kaggle was imported into Tableau Desktop for preprocessing and visualization. Interactive dashboards and Tableau Stories were created to understand ordering patterns, meal preferences, pricing sensitivity, promotional impact, customer demographics, seasonal demand, and delivery preferences. The completed dashboard and story were published to Tableau Public and embedded into a responsive Bootstrap website developed using HTML5, CSS3 and Visual Studio Code. This project demonstrates how Business Intelligence tools transform raw food ordering data into meaningful consumer insights that help restaurants and delivery platforms optimize their strategies.

INTRODUCTION

Business Intelligence helps organizations make informed decisions by converting raw data into useful information. In this project Tableau Desktop was used to analyze a food delivery dataset and understand how pricing, promotions, order timing, customer demographics, and delivery preferences influence consumer choices and ordering habits. The project covers the complete BI workflow including data collection, preprocessing, visualization, dashboard creation, story development, publishing and website integration.

PROBLEM STATEMENT

Food delivery platforms and restaurants must understand consumer behavior to remain competitive in a rapidly growing market. Factors such as meal preferences, order timing, price sensitivity, promotional offers, customer demographics, and delivery duration all influence purchasing decisions. Manual analysis is difficult because of the large amount of transactional data. This project uses Tableau to create dashboards and stories that help food businesses make data-driven decisions to improve customer retention and optimize marketing efforts.

OBJECTIVES

The objectives are to collect a food delivery dataset from Kaggle, connect it with Tableau Desktop, preprocess and validate the data, create meaningful visualizations, build an interactive dashboard, create a Tableau Story, publish the workbook to Tableau Public and integrate it into a Bootstrap website using HTML5, CSS3 and Visual Studio Code.

SCOPE OF THE PROJECT

The project focuses on analyzing historical food ordering data to identify ordering trends, pricing strategies, promotional effectiveness, customer preferences, and delivery performance. Interactive dashboards help managers explore data through filters and improve consumer targeting and operational decisions. The same workflow can be extended to real-time business environments.

DATASET COLLECTION

The dataset was downloaded from Kaggle in CSV format. It contains approximately 1000 order records with attributes including Order ID, Customer Demographics, Meal Type, Order Time, Day of Week, Product Price, Delivery Fee, Promotion Status, Order Value, Customer Rating, Delivery Time, and Order Frequency. The CSV file was imported into Tableau Desktop using the Text File connector.

DATASET DESCRIPTION

Field Name	Description	Data Type
Order ID	Unique identifier for each order	String
Customer Demographics	Age group or customer segment	String
Meal Type	Cuisine or meal category ordered	String
Order Time	Time of day the order was placed	String/Time
Day of Week	Day when the order was placed	String
Product Price	Price of the ordered item(s)	Number (Float)
Delivery Fee	Fee charged for delivery	Number (Float)
Promotion Status	Whether a promotion was applied	String (Yes/No)
Order Value	Total value of the order	Number (Float)
Customer Rating	Rating given by the customer	Number (Float)
Delivery Time	Time taken for delivery in minutes	Number (Integer)
Order Frequency	Number of orders placed by the customer	Number (Integer)

SOFTWARE, TOOLS AND TECHNOLOGIES USED

Kaggle was used for dataset collection. Tableau Desktop was used for preprocessing, visualization, dashboards and stories. Tableau Public was used for publishing. Visual Studio Code, HTML5, CSS3 and Bootstrap 5 were used to develop the website. Google Chrome was used for testing and Google Drive for sharing the demonstration video.

CONNECTING THE DATASET TO TABLEAU DESKTOP

The food delivery dataset downloaded from Kaggle was imported into Tableau Desktop using the Text File connector. After selecting the CSV file, Tableau

displayed the Data Source page where all fields and records were loaded automatically. Around 1000 records were successfully imported. This step ensured that the dataset was ready for preprocessing and visualization.

DATA REVIEW AND EXPLORATION

After importing the dataset, each field was reviewed to understand its structure and business meaning. Tableau automatically categorized fields into Dimensions and Measures based on their data types. Customer Demographics, Meal Type, Order Time, Day of Week, and Promotion Status were identified as dimensions, while Product Price, Delivery Fee, Order Value, Customer Rating, Delivery Time, and Order Frequency were measures.

DATA CLEANING AND PREPARATION

Basic preprocessing was performed to ensure data quality. Missing values were checked, and duplicate records were removed. Delivery Time values that were unreasonably high (outliers) were reviewed and corrected where necessary. Numeric fields such as Product Price and Delivery Fee were verified for consistency.

CALCULATED FIELDS

- **Price Sensitivity Index:** Compared Product Price to the average price for that Meal Type.
- **Promotion Effectiveness:** Measured the difference in Order Value between promoted and non-promoted orders.
- **Order Time Slot:** Categorized Order Time into segments like Breakfast, Lunch, Dinner, and Late Night.
- **Average Rating per Meal:** Calculated the average Customer Rating for each Meal Type.

VISUALIZATIONS

Visualization 1 - Order Volume by Meal Type (Bar Chart)

Highlights cuisines generating the highest volume. Supports menu planning and inventory decisions.

Visualization 2 - Promotion Effectiveness (Stacked Bar Chart)

Compares promoted vs. non-promoted orders to evaluate marketing ROI.

Visualization 3 - Order Timing Distribution (Donut Chart)

Displays distribution across Breakfast, Lunch, Dinner, and Late Night slots.

Visualization 4 - Price vs. Customer Rating (Scatter Plot)

Supports pricing strategy decisions by balancing cost and customer satisfaction.

Visualization 5 - Seasonal Demand Analysis (Line Chart)

Displays fluctuations in demand over seasonal periods for inventory planning.

Visualization 6 - Sales by Customer Demographics (Pie Chart)

Identifies key customer segments (Gen Z, Millennials, etc.) for targeted marketing.

Visualization 7 - Delivery Time vs. Customer Rating (Bubble Chart)

Demonstrates how logistics speed influences satisfaction ratings.

Visualization 8 - Order Heat Map by Day and Time (Heat Map)

Highlights peak ordering periods (e.g., Friday Dinner) for operational planning.

DASHBOARD AND STORY DEVELOPMENT

After creating the visualizations, they were combined into an interactive Dashboard with filters (Meal Type, Time Slot, Day, etc.). A Tableau Story was also developed to present the analysis in a sequential format, explaining personas, promotional impact, and delivery trends.

KEY FINDINGS

The analysis showed that Meal Type significantly influences order frequency. Promotional campaigns positively affected volume, particularly for Asian cuisine. Price sensitivity analysis revealed a priority for value-for-money. Delivery time directly impacted ratings. Millennials were identified as the largest segment, while Gen Z preferred late-night orders.

CONCLUSION

The project successfully demonstrated the complete Business Intelligence lifecycle. Tableau transformed raw food ordering data into meaningful consumer insights. The Bootstrap website provided a professional platform for presenting the completed project online.

FUTURE SCOPE

Future enhancements may include integrating real-time databases, applying machine learning for volume prediction, and developing mobile applications for real-time monitoring of restaurant performance.